

Cortisol Dysregulation and Anxiety Levels Among University Students: A Comparative Study of Baseline Evidence and Primary Data. – by, Wadha Abdullah Almansouri.

Abstract

University students experience high levels of anxiety, which may be linked to biological stress mechanisms. Cortisol, regulated by the hypothalamic–pituitary–adrenal (HPA) axis, plays a key role in stress response. This study examines the relationship between cortisol levels and anxiety using a quantitative cross-sectional design. Salivary cortisol and GAD-7 scores will be analysed using correlation and regression. The study aims to determine whether cortisol dysregulation predicts anxiety among students.

Introduction

Anxiety is a common mental health issue among university students, often driven by academic and lifestyle pressures (Beiter et al., 2015). Beyond psychological explanations, research highlights the role of the HPA axis in regulating stress responses through cortisol (McEwen, 2007). Under normal conditions, cortisol follows a diurnal rhythm; however, chronic stress can disrupt this pattern, leading to dysregulation (Adam et al., 2017). Such dysregulation has been linked to anxiety and other mental health outcomes (Chida & Steptoe, 2009). Despite this, limited research has examined this relationship in real student settings using physiological measures, creating a clear research gap.

 **Aim**

To examine the relationship between **cortisol levels and anxiety** among university students.

 **Hypotheses**

- H1:** Higher cortisol → Higher anxiety
- H2:** Cortisol predicts anxiety levels
- H3:** Findings align with existing literature

 **Ethics**

- ✓ Informed consent obtained
- ✓ Participation is voluntary
- ✓ Data is anonymous & confidential
- ✓ Secure data storage
- ✓ Psychological support provided
- ✓ Right to withdraw anytime

 **Limitations**

- Cortisol influenced by sleep, caffeine, activity
- Cross-sectional → no causality
- Small sample size
- Context-specific findings

 **Methodology**

 **Participants**

- 60–80 university students
- Age 18+
- Voluntary participation

 **Measures**

- Salivary Cortisol**
 - Collected ~30 mins after waking
- GAD-7 Scale**
 - 7-item anxiety questionnaire

 **Design**

- Quantitative
- Cross-sectional
- Correlational

 **Analysis**

- Descriptive statistics
- Pearson correlation
- Linear regression ($p < .05$)

References

Adam, E. K., Quinn, M. E., Tavernier, R., McQuillan, M. T., Dahlke, K. A., & Gilbert, K. E. (2017). Diurnal cortisol slopes and mental health outcomes: A systematic review and meta-analysis. *Psychoneuroendocrinology*, *83*, 25–41.

Beiter, R., Nash, R., McCrady, M., Rhoades, D., Linscomb, M., Clarahan, M., & Sammut, S. (2015). The prevalence and correlates of depression, anxiety, and stress in a sample of college students. *Journal of Affective Disorders*, *173*, 90–96.

Chida, Y., & Steptoe, A. (2009). Cortisol awakening response and psychosocial factors: A systematic review and meta-analysis. *Biological Psychology*, *80*(3), 265–278.

McEwen, B. S. (2007). Physiology and neurobiology of stress and adaptation: Central role of the brain. *Physiological Reviews*, *87*(3), 873–904.

Spitzer, R. L., Kroenke, K., Williams, J. B. W., & Löwe, B. (2006). A brief measure for assessing generalised anxiety disorder: The GAD-7. *Archives of Internal Medicine*, *166*(10), 1092–1097.